INCH-POUND

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MILITARY SPECIFICATION

PUMP, FUEL, ELECTRICAL: 12 AND 24 VOLT DC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electrically operated, in-line fuel pumps for use with 12 and 24 volt (V), direct current (dc) automotive electrical systems.

1.2 Classification. Pumps shall be furnished in the following types and classifications according to drawings as specified (see 6.2 and 6.4).

Type I	- Standard duty (630 hours)
Type II	- Long life (5040 hours)
Grade 1	- 12 V dc nominal
Grade 2	- 24 V dc nominal
Class 1	- Negative ground
class 2	- Positive ground

Benefits comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specification and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

VV-F-800	- Fuel oil, Diesel.
VV-G-1690	- Gasoline, Automotive, Loaded or Unleaded.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-F-13927	- Fungus Resistance Test, Automotive Components.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-461A, NOTICE 4	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462, NOTICE 3	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-45662	- Calibration Systems Requirements.
MS51321	- Pump, Fuel Electrical: 24 Volts dc, 25 gph Capacity -

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks am available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

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2.2 **Other publications.** The following document forms a part of this specification to the extent-specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D235	- Mineral Spirits (Petroleum Spirits) (Hydrocarbon Dry Cleaning Solvent) Specification For.
ASTM B117	- Salt Spray (Fog) Testing, Method of
ASTM D3951	- Commercial Packaging, Practice For.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedende. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **Qualification.** Pumps furnished under this specification shall be products which are qualified for listing on the applicable qualified products list (QPL) at the time set for opening of bids (see 4.4 and 6.3).

3.2 Materials. Materials shall be as specified herein and in referenced specifications and drawings. Materials shall be uniform and free of defects and imperfections that might affect the serviceability and appearance of the finished product (see 4.8.1).

3.2.1 **Recycled, virgin and reclaimed materials.** There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 6.5).

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3.3 Design and construction. All pumps shall conform to MS51321 (see 4.8.1, 4.8.2, 6.2 and 6.4).

3.3.1 Filter. The pump shall incorporate a filter to protect the fuel supply system. The filter shall be freely removable for cleaning (see 4-8.1).

3-4 Performance

3.4.1 Output. The pump output for both grades and types shall equal or exceed the performance curves specified on MS51321 the pump shall operate at a nominal voltage of 12 or 24 V dc within the voltage range listed in table I. During testing, the load current shall not exceed 1.75 amperes. The test fuel shall conform to one of the following: ASTM D235, type I; TT-T-291. type I; P-D-680, type I; VV-F-800, MIL-G-3056 or MIL-T-5624 (see 4.8.3).

TABLE I. Voltage.

Grades	Nominal voltage	Minimum voltage	Maximum voltage
1	12	6	18
2	24	18	30

3.4.2 Dielectric strength. When operating the pump with the internal ground and the radio suppressor ground strap disconnected, insulation shall show no loosening, cracking, charring, burning or smoking {see 4.8.4).

3.4.3 Electromagnetic interference suppression. The pump shall meet the requirements of MIL-STD-461A, notice 4, methods RE02 and CE04 (see 4.8.5).

3.4.4 Fuel resistance. After being tested, the pump shall evidence no damage or leakage and shall subsequently meet the performance requirements of 3.4.1 (see 4-8.6).

3.4.5 Environmental.

3.4.5.1 Extreme temperature resistance. Pump fuel flow, at any given pressure, shall be not less than 67 percent (%) of the value reflected by the performance curves of MS51321, in ambient temperatures ranging from minus 65 to plus 160 degrees Fahrenheit (°F) (see 4.8.7.1).

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3.4.5.2 **Fungus resistance.** After testing, the pump shall evidence no damage and shall subsequently meet performance requirements of 3.4.1 (see 4.8.7.2).

3.4.5.3 Vibration resistance. The pump shall evidence no damage or leakage and shall subsequently meet the requirement of 3.4.1 when subjected to a simple harmonic motion amplitude of 0.03 inch at a frequency range from 10 to 55 Hertz (Hz) and return to 10 Hz in approximately 1 minute. This motion shall be applied for a period of 2 hours in each of 3 mutually perpendicular axes (total 6 hours) (see 4.8.7.3).

3.4.5.4 **Shock resistance.** The pump shall evidence no damage or leakage and shall meet the requirements of 3.4.1 when subjected to sawtooth wave shocks of 25 gravity units (g) at 18 milliseconds (ms) along three mutually perpendicular axes (see 4.8.7.4 and 4.8.7.4.1).

3.4.5.5 **Corrosion resistance.** After cyclic exposure to a salt spray atmosphere, the pump shall meet the requirements of 3.4.1 (see 4.8.7.5)

3.4.6 Endurance. The pump shall maintain a delivery versus pressure relationship not leas than that specified on MS51321 for a period described below (see 4.8.8):

Type I - 630 hours
Type II - 5040 hours

3.5 Marking. Pump shall be marked in accordance with MIL-STD-130 and shall include the following information: Polarity, voltage, Military part number, and manufacturer-s identification (see 4.8.2).

3.6 Workmanship. Workmanship shall be such as to assure a product free of chips, burrs, sharp edges, rust, loose or defective connectors, or faulty soldering and shall not compromise, limit or reduce the capability of the pump in the performance of its intended use (see 4.8.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 **Responsibility for compliance.** All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor-s overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformancee does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 **Inspection equipment.** Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10% of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspection.

a. Qualification inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
   1. Examination (see 4.5.2).
   2. Tests (see 4.5.3).
c. Control tests (see 4.6).

4.3 **Inspection conditions.** Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature: 75 ± 18°F
b. Barometric pressure: 28.5 + 2 inches mercury
   - 3
c. Relative humidity: 50 ± 30%

4.4 **Qualification inspection.** For the purpose of qualification (see 3.1) the prospective supplier shall submit seven pumps. These specimens, properly marked with identifying information, shall be representative of the units proposed to be furnished to the Government in connection with future procurement. Qualification testing shall be conducted by the Government at a place designated by the Government. The pumps shall be inspected as specified in table II and tested in the order listed in table III.

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TABLE II. Classification of inspections.

Title	Requirement	Inspection	Qualification	Quality conformance		
				Examination	Tests	Control
Materials and construction	3.2, 3.3, 3.3.1	4.8.1	X			
Defects (see 4.8.2 and table III)	3.3, 3.5 and 3.6	4.8.2	X	X		
Output	3.4.1	4.8.3	X		X	X
Dielectric strength	3.4.2	4.8.4	X			X
Electromagnetic interference suppression	3.4.3	4.8.5	X			
Fuel resistance	3.4.4	4.8.6	X			
Extreme temperature resistance	3.4.5.1	4.8.7.1	X			
Fungus resistance	3.4.5.2	4.8.7.2	X			
Vibration resistance	3.4.5.3	4.8.7.3	X			X
Shock resistance	3.4.5.4	4.8.7.4	X			X
Corrosion resistance	3.4.5.5	4.8.7.5	X			
Endurance	3.4.6	4.8.8	X			

TABLE III. Qualification tests.

Sample No.	Test	Paragraph No.
1	Output	4.8.3
	Endurance	4.8.8
	Output	4.8.3
2	Output	4.8.3
	Dielectric strength	4.8.4
3	Output	4.8.3
	Electromagnetic interference suppression	4.8.5

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TABLE III. Qualification tests - Continued.

Sample No.	Test	Paragraph No.
4	Output	4.8.3
	Extreme temperature resistance	4.8.7.1
	Output	4.8.3
5	Output	4.8.3
	Fuel resistance	4.8.6
	Output	4.8.3
6	Output	4.8.3
	Vibration resistance	4.8.7.3
	Output	4.8.3
	Fungus resistance	4.8.7.2
	Output	4.8.3
7	Output	4.8.3
	Shock resistance	4.8.7.4
	Output	4.8.3
	Corrosion resistance	4.8.7.5
	Output	4.8.3

4.4.1 **Retention of qualification.** Certification shall be requested every 2 years from each manufacturer listed on the QPL to retain listing on the QPL. This certification shall be forwarded to the preparing activity and shall be signed by a responsible official of management, attesting that the listed product still meets the requirements of the current issue of the specification, is available from the listed plant, and can be produced under the same conditions as originally qualified; i.e., same process, materials, construction, design, manufacturer's part number or designation. Failure to provide certification shall be cause for removal from the QPL.

4.5 Quality conformance inspection.

4.5.1 **Sampling.**

4.5.1.1 Lot formation. An inspection lot shall consist of all the fuel pumps of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

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4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.2 Examination.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL) (see 6.3).

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 **Classification of defects.** For examination purposes, defects shall be classified as listed in table-IV.

TABLE IV. **Classification of defects.**

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Assembly, incomplete (see 3.3).	Visual
102	Dimensions affecting interchangeability, out of tolerance (see 3.3).	SIE 1/
103	Faulty workmanship affecting performance (see 3.6).	Visual
Minor	AQL 2.5% Defective	
201	Dimensions not affecting interchangeability out of tolerance (see 3.3).	SIE
202	Identification marking, improper (see 3.5).	Visual
203	Faulty workmanship affecting appearance (see 3.6).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Tests. Samples selected in accordance with 4.5.1.2 shall be subjected to the quality conformance tests specified in table II, and shall conform to an AQL of 6.5 on the basis of percent defective-

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4.6 Control tests. Control teats shall be conducted on 3 pump assemblies from each lot of 500 units consecutively produced, except that not more than 6 nor less than 3 units shall be selected in any 30-day period. The control teat samples shall be examined for the defects s cified in 4.5.2 and subsequently subjected to the tests specified in table II in the order listed in table V.

TABLE V. Control test sequence.

Sample No.	Test	Paragraph No.
1	Output test	4.8.3
	Dielectric strength test	4.8.4
2	Output test	4.8.3
	Vibration resistance test	4.8.7.3
	Output test	4.8.3
3	Output test	4.8.3
	Shock resistance test	4.8.7.4
	Output test	4.8.3

4.7 Failure. Failure of any pump assembly to pass any of the specified inspections shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.8 Methods of inspection.

4.8.1 Materials and constructior. Conformance to 3.2, 3.3, 3.3.1 through 3.3.4 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.8.2 Defects. Conformance to 3.3, 3.5 and 3.6 shall be determined by examination for the defects listed in table IV. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

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4.8.3 Output. To determine conformance to 3.4.1, the pump shall be tested for output at the minimum and maximum voltages listed in table I for the grade of pump under test unless otherwise specified herein the test fuel shall be gasoline conforming to MIL-G-3056 or ASTM-D-235. The pump inlet and the test fuel level shall be maintained at the same height within plus or minus two inches if the pump fails. A minimum of five values each of discharge pressure, volume, and current consumption shall be obtained at each voltage setting. The readings shall be taken at points approximately equidistant between pressure at zero delivery and delivery at zero pressure. All pressure reading shall be taken at the pump outlet.

4.8.4 Dielectric strength. To determine conformance to 3.4.2, the pump shall be disassembled or cut open to the extent necessary to permit the internal ground connection to be disconnected or isolated, and to permit observation of the internal wiring. The pump shall then be subjected to 500 V root mean square (rms) at a frequency of 60 cycles per second (cps) applied between each conductor and ground for a period of 1 minute.

4.8.5 **Electromagnetic interference suppression.** To determine conformance to 3.4.3, the pump shall be tested in accordance with methods CE04 and RE02 of MIL-STD-462, Notice 3.

4.8.6 Fuel resistance To determine conformance to 3.4.4, pumps shall be subjected to the fuel resistance tests specified in table VI. Except for the first 48 hours of phase I and phase III testing, pumps shall be operated at rated flow for 2 hours per day

TABLE VI. **Fuel resistance schedule.**

Phase	Period	Medium	Temperature °F + 5°
Phase I soak	168 hours	TT-S-735, type III	75
Phase II dry	4 hours	Air	155
Phase III soak	168 hours	TT-S-735, type III	75
Phase IV dry	4 hours	Air	155

4.8.7 Environmental.

4.8.7.1 Extreme temperature resistance. To determine conformance to 3.4.5.1, the pump shall be mounted and operated as specified in 4.8.3, except that the values shall be recorded at minus 63 + 2°F and plus 158 + 2°F and the pump shall not be reoperated at 18 V at the higher temperature. The fuel used shall conform to MIL-G-3056, type I, for plus 158°F operation and MIL-G-3056 type II for minus 63°F operation.

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4.8.7.2 **Fungus resistance.** To determine conformance to 3.4.5.2, pumps shall be subjected to the fungus resistance test specified in MIL-F-13927 for class 2, method B incubation. The period of incubation shall be a continuous 90 days.

4.8.7.3 **Vibration resistance.** To determine conformance to 3.4.5.3, the pump shall be mounted as in intended use using suitable mounting apparatus to assure that mounting is free from resonances over the test frequency range. The specimens shall be subjected to a simple harmonic motion having an amplitude of 0.03 inch (0.06 inch maximum total excursion), the frequency being varied uniformly between the approximate limits of 10 and 55 Hz. The entire frequency range, from 10 to 55 Hz and return to 10 Hz, shall be traversed in approximately 1 minute. This motion shall be applied for a period of 2 hours in each of 3 mutually perpendicular directions (total of 6 hours). At the end of test the pump shall conform to 3.4.1.

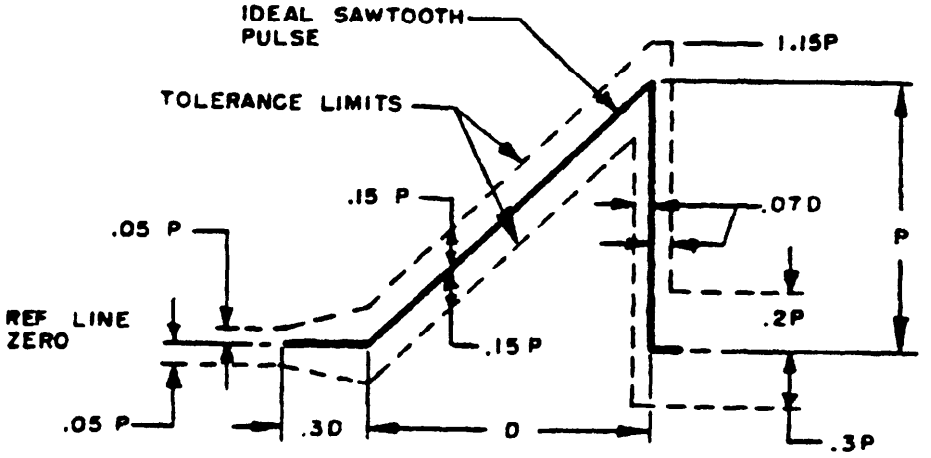


FIGURE 1. Terminal-peak sawtooth shock pulse configuration.

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4.8.7.4 Shock resistance. To determine conformance to 3.4.5.4, the pump shall be mounted on vibration isolators and equipment racks as in intended use. The isolators shall be functional during the test. The assembly is subjected to three sawtooth shock pulses of 25 g at 18 ms duration in each direction along 3 mutually perpendicular axes of the pump (total of 18 shocks). The shock pulses shape shall be in accordance with figure 1 of amplitude and time duration as specified. At the end of the test, the pump shall be operated and shall meet the requirements of 3.4.1.

4.8.7.4.1 Shock, optional test procedure

4.8.7.4.1.1 Apparatus. The shock-testing apparatus herein, as shown in figures 2 through 9 inclusive, shall be used. The main structure consists of a cast-iron base to which upright channels are securely bolted. The tops of the upright channels are secured to a top channel which also supports rods to guide the moveable carriage assembly. The other ends of the guide rods are screwed into the base. The carriage assembly carries on its underside a stiff calibrated spring resiliently mounted, but at the same time having helical compression springs sufficiently strong to hold the calibrated spring in contact with the effective cylindrical bearings. The cast-iron base shall be single piece constructed so that it has no practical yielding. The prime requisite in the construction of the moveable carriage assembly is rigidity. For this reason, 1/4 inch thick steel shall be used. After being securely fastened together with screws, the carriage parts shall be welded so as to give the equivalent of single-piece construction.

4.8.7.4.1.2 Mounting accessories. Mounting accessories, which will assure rigid fastening of specimens to the carriage assembly, shall be used. Since special tests may require special mounting fixtures, the carriage test plate shall be provided with a number of extra holes. The top of the carriage frame shall be provided with a holding bolt to facilitate the rigid clamping necessary.

4.8.7.4.1.3 **Stationary anvil.** The stationary anvil shall be made of hardened tool steel with a radius-of curvature of 1 1/2 inches and shall be bolted tightly to the base.

4.8.7.4.1.4 Calibrated spring. The calibrated spring shall be made of first-quality carbon tool steel, spring-tempered, and having a hardness of 85 to 95, Rockwell B scale. It shall be annealed and ground flat stock of the following composition:

	Percent
Carbon	1.08 to 1.18
Manganese	.20 to .30
Silicon	.20 to .30
Phosphorous and sulfur	.025 maximum

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The spring shall be 1 3/4 inches wide and approximately 0.150 inch thick, and shall have an overall length of 7 1/2 inches. The approximate spring constant (K) shall be between 5,000 and 5,500 pounds-per-inch deflection.

4.8.7 .4.1.5 Calibration. The actual calibration shall be made on the completely assembled machine. Static loads up to 500 pounds shall be applied to the top of the carriage frame, and the spring deflections measured by means of a dial gage or height gage. The gage shall be mounted on the top surface of the cast-iron base as one reference surface, and measurements shall be taken at the center of the carriage test plate. The spring constant shall be determined as follows:

$$\text{Spring constant (K)} = \frac{\text{load in pounds}}{\text{inches in deflection}}$$

For a spring constant of 5,000, the deflection for a 500-pound load would be 0.100 inch.

4.8.7.4.1.6 Calibration curves. Use of the shock-testing mechanism is facilitated by plotting calibration curves for various weights of specimens. Ordinarily, parallel curves for specimen weights of 0.5 pounds and 2.0 pounds will be adequate, and intermediate weights can be estimated. Figure 8 shows such curves arbitrarily calculated on the basis of a mean spring constant (K) of 5,250 and a total carriage-and-mounting-fixture weight of 9 3/4 pounds.

4.8.7.4.1.7 Computation of acceleration (G). The following formula for obtaining the required acceleration or shock, in-terms of gravity units (G), is utilized for variations of component weight and spring constant (K):

$$G = \sqrt{\frac{2hK}{w}} \quad \text{inches}$$

When G is specified, the necessary height may be determined from:

$$h = \frac{G^2 w}{2K} \quad \text{inches}$$

Where:

G = nominal value of acceleration (gravity units).
h = height from which carriage is dropped (inches).
K = spring constant (pounds-per-inch deflection).
w = weight of carriage and its test load (pounds).

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4.8.7.4.1.8 Procedure. The shock-testing apparatus shall be mounted on a sturdy laboratory table or equivalent base and leveled before use. The specimen shall be rigidly mounted on the carriage. For each blow, the carriage shall be raised to the height necessary to obtain the required acceleration and then allowed to fall. Means shall be provided to prevent the carriage from striking the anvil a second time.

4.8.7.5 Corrosion resistance. To determine conformance to 3.4.5.5, the pump shall be subjected to the 20% salt spray test of ASTM B117 for a duration of 100 hours. After salt spray testing the pump shall conform to 3.4.1.

4.8.8 Endurance. To determine conformance to 3.4.6, the pump shall be operated as specified in table VII. The test fuel shall be diesel oil, grade DF-2, conforming to VV-F-800.

TABLE VII. Endurance.

Voltage grades		Delivery per hour		Time-hours	
1	2	Gallons	Liters	Type I	Type II
6	18	0	0	15	120
		3	11.5	150	1200
		15	57	150	1200
18	30	0	0	15	120
		3	11.5	150	1200
		25	95	150	1200

4.9 Packaging and packing.

4.9.1 Army packaging and packing (see 5.1).

4.9.2 Other activities.

4.9.2.1 **Lot formation.** A lot shall consist of all packs, packed in accordance with one level (see 5.1.1 and 5.2.3) from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.9.2.2 Examination

4.9.2.2.1 Sampling for examination. Samples for the examination listed in table VI shall be selected in accordance with inspection level II of MIL-STD-105.

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4.9.2.2.2 **Acceptable quality level.** Each sample selected in accordance with 4.9.2.2.1 shall be examined to determine conformance to the following acceptable quality levels (AQL's) on the basis of % defective:

Classification	AQL
Major	1.0
Minor	2.5

4.9.2.2.3 **Classification of defects.** For examination purposes, defects shall be classified as listed in table VIII.

TABLE VIII. **Classification of defects**.

Category	Defect	Method of examination
Major	AQL 1.0% Defective	
101	Illegible marking (see 5.1.1 and 5.2.4).	Visual
102	Incorrect marking (see 5.1.1 and 5.2.4).	Visual
Minor	AQL 2.5% Defective	
201	Improper closure and strapping of exterior container (see 5.1.1 and 5.2.3).	Visual

4.9.2.3 **Control test.**

4.9.2.3.1 **Selection of samples.** Control test samples shall be selected at the rate of 3 of each 500 units produced except that not more than 6 nor less than 3 shall be selected in any 30-day period.

4.9.2.3.2 **Testing and examination for packages, and packs.** Packages and packs selected in accordance with 4.9.2.3.1 shall be examined for the defects specified in 4.9.2.2.3. Packages shall subsequently be subjected to the test specified in 4.9.2.4.

4.9.2.4 **Packaging test.** To determine conformance to 5.2.1.4, the unit package shall be subject to the tests specified for method IC-2 of MIL-P-116.

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5. PACKAGING

5.1 For Army use.

5.1.1 **Preservation, packaging, packing, and marking.** Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

5.2 Other activities.

5.2.1 Preservation-packaging. Preservation-packaging shall be level A or commercial, as specified (see 6.2).

5.2.1.1 Level A.

5.2.1.2 Cleaning. The pump shall be cleaned in accordance with process C-1 of MIL-P-116.

5.2.1.3 Drying. The pumps shall be dried in accordance with any applicable procedure of MIL-P-116.

5.2-1.4 **Unit packaging.** Immediately after cleaning and drying, all openings shall be closed with non-corrosive caps or plugs. The connecting cable shall be wrapped around the body of the pump or placed in a manner to reduce the cube without kinking the cable. The pumps shall be individually unit packaged in accordance with submethod IC-2 of MIL-P-116. When tested as specified in 4.9.2.4, the package shall meet the requirements for method IC-2 of MIL-P-116.

5.2.1.5 **Intermediate packing**. Four unit packages shall be intermediate packaged into a class weather resistant, style RSC, Type CF fiberboard box conforming to PPP-B-636. Closure shall be in accordance with the box specification-

5.2.2 **Commercial.** Cleaning, drying and packaging shall be in accordance with ASTM D3951.

5.2.3 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2-3.1 Level A. Pumps packaged as specified in 5.2.1 shall be packed in overseas type, style optional, grade B, wood boxes conforming to PPP-B-601, or PPP-B-621. Closure and strapping shall be in accordance with the box specification.

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5.2.3.2 Level B. Pumps, packaged as specified in 5.2.1, shall be packed as specified in 5.2.3.1, except the containers may be domestic; or packed in class weather resistant, style optional, fiberboard boxes conforming to PPP-B-636.

5.2.3.3 **Commercial.** Packing shall be in accordance with ASTM D3951.

5.2.4 Marking. In addition to any special marking (see 6.2) required by the contract or order, unit packages, intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Type I fuel pumps furnished under this specification are intended for use in military motor vehicles or other military equipment (personnel heaters and winterization kits, for example) requiring fuel distribution. Type II fuel pumps are intended primarily for use in engine driven portable electric generator sets where long life is a requirement.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type, grade, class, and applicable drawing or MS standard (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
d. Title, number, and date of applicable drawings (see 3.3).
e. If responsibility for inspection shall be other than as specified (see 4.1).
f. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).
g. If inspection conditions shall be other than as specified (see 4.3).
h. Selection of applicable level and packaging requirements (see 5.1, 5.2.1 and 5.2.3).
i. If any special marking shall be required (see 5.2.4).
j. Fuel shall be other than as specified (see 4.8.3).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion of the applicable QPL, whether or not such products have actually been so listed by that date. The attention of contractors is called to this requirement, and manufacturers are urged to have the products that they propose to offer to the Federal

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Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the parts covered by this specification. The activity responsible for the QPL is the Commanding General, US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, and information pertaining to qualification of products may be obtained from that activity.

6.4 **National Stock Number** (NSN) superseded NSN's are included in table IX for reference.

TABLE IX. **Part number cross-reference**.

New Government description	NSN	Previous Gov't part number	Previous drawing no.
MS51321-1-12N1	2910-00-735-4013 2910-00-915-4818	8376298	7354062
MS51321-1-24N1	2910-00-710-6054 2910-00-735-4014 2910-00-933-6104	7392887 7354014 7354062 8376299	7392887 7354062 7354062 7354062
MS51321-1-24P1	4320-00-837-7470	7620097	7620097
MS51321-2-12N1	2910-00-847-9250		
MS51321-2-24N1	2910-00-930-9367 2910-00-073-0165 2910-00-676-4957	10910278 7392887 7748814	7392887 7392887 7392887
MS51321-3-24N1	2910-00-679-9743	10861299	10861299

6.5 **Definitions**.

6.5.1 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.5.2).

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6.5.2 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, li uid, semisolid, or contained gaseous material resulting from industrial commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.6 Subject term (key word) listing.

Corrosion resistance
Electromagnetic interference suppression
Extreme (high and low) temperature resistance
Fuels
Fungus resistance
Marking per MIL-STD-130
Packaging per MIL-P-116
Qualification
Recovered material
Shock resistance
Vibration resistance

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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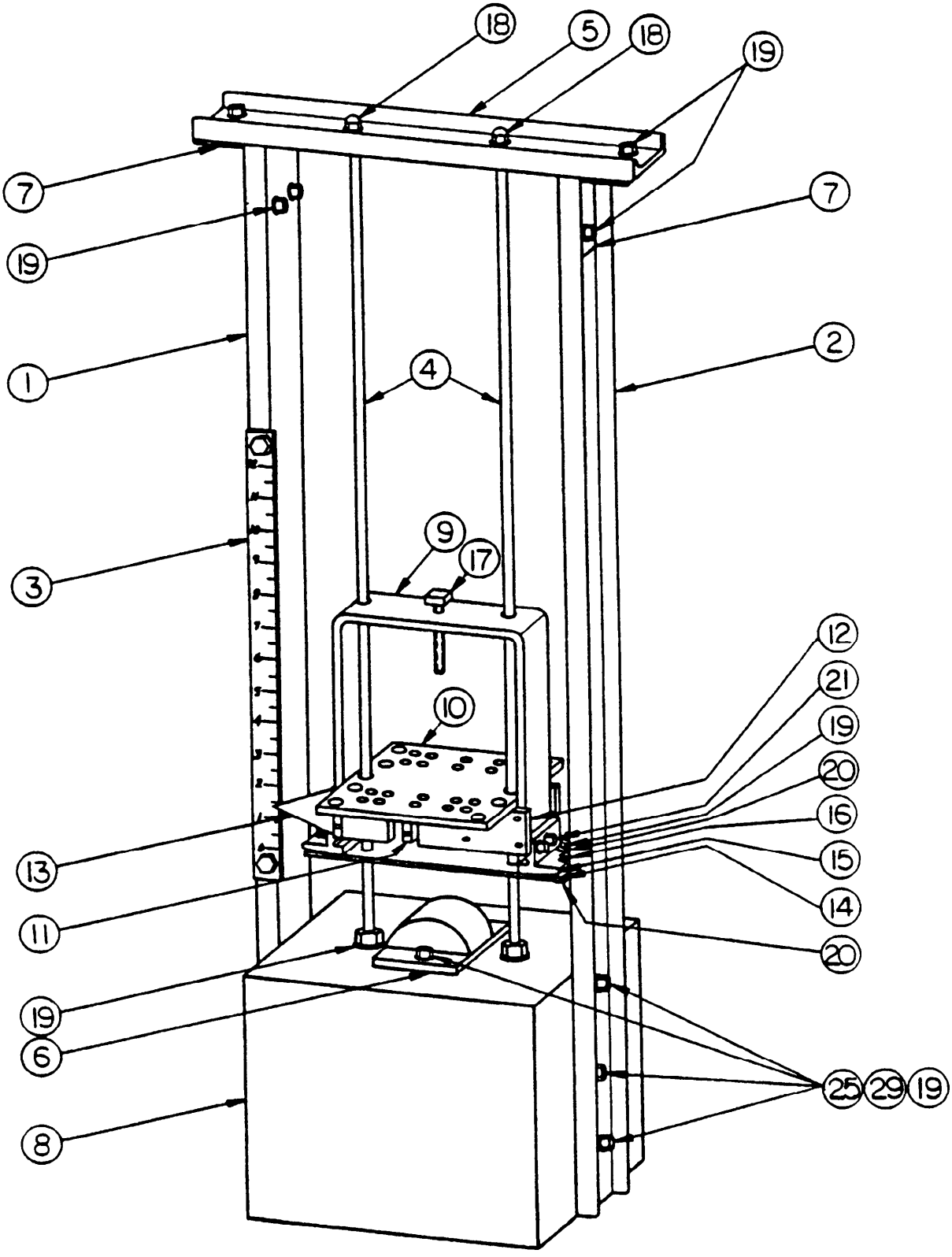


FIGURE 2. General assembly of shock-testing apparatus (See figure 8 for notes and bill of materials).

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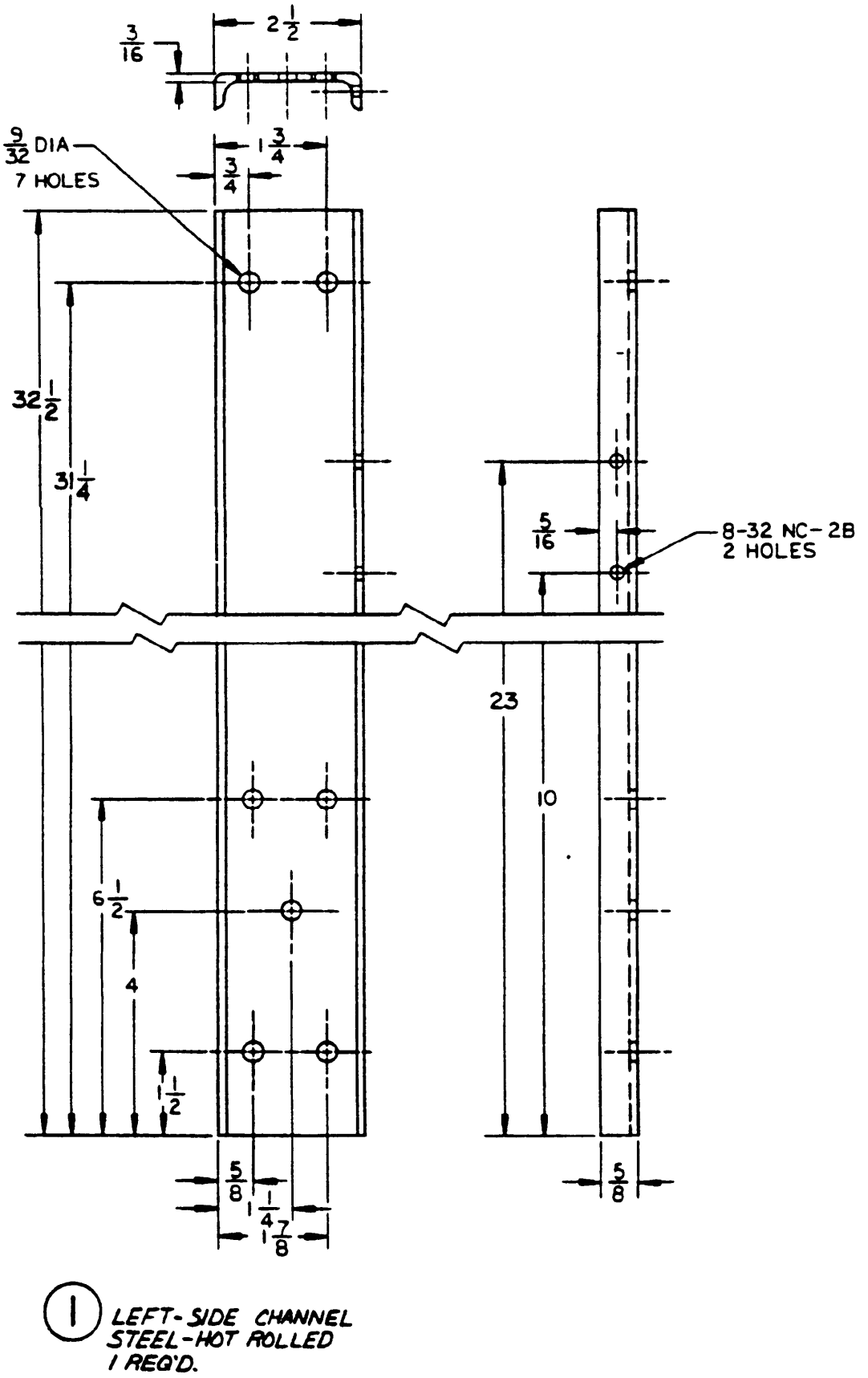
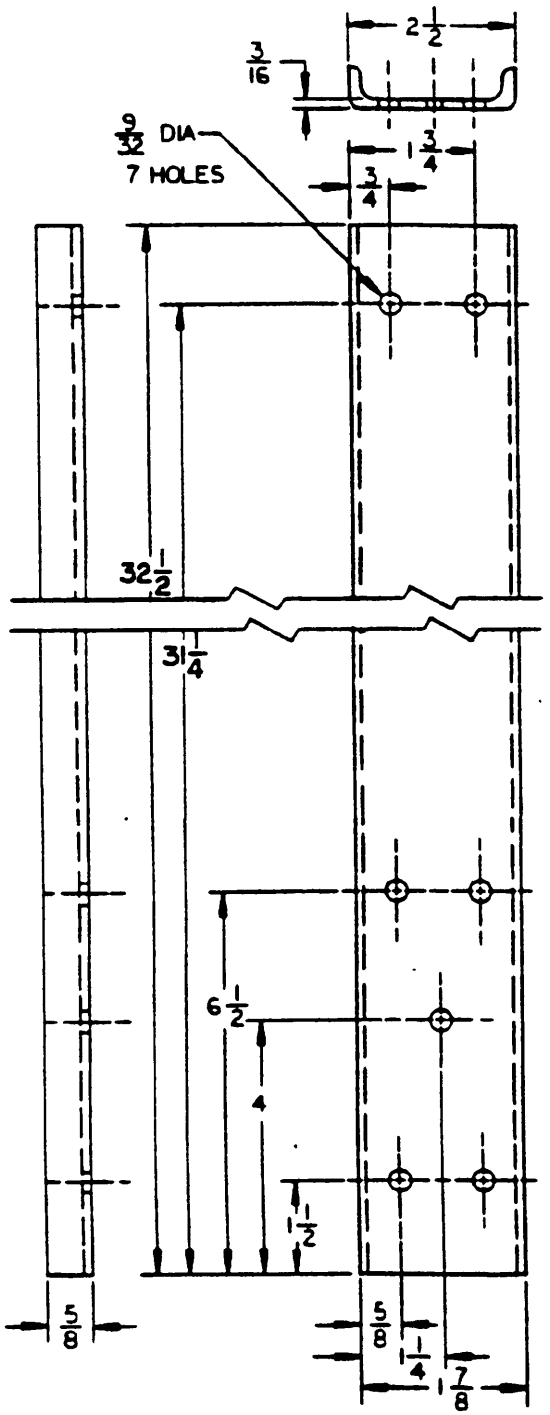
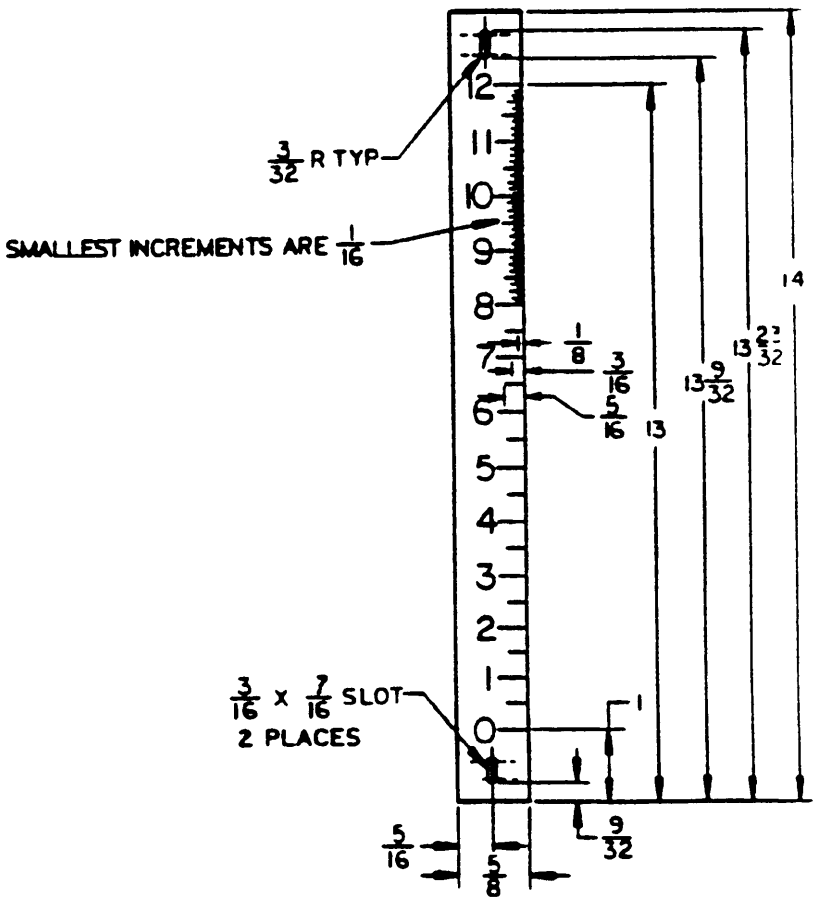


FIGURE 3. Main frame details (See figure 8 for notes and bill of materials).

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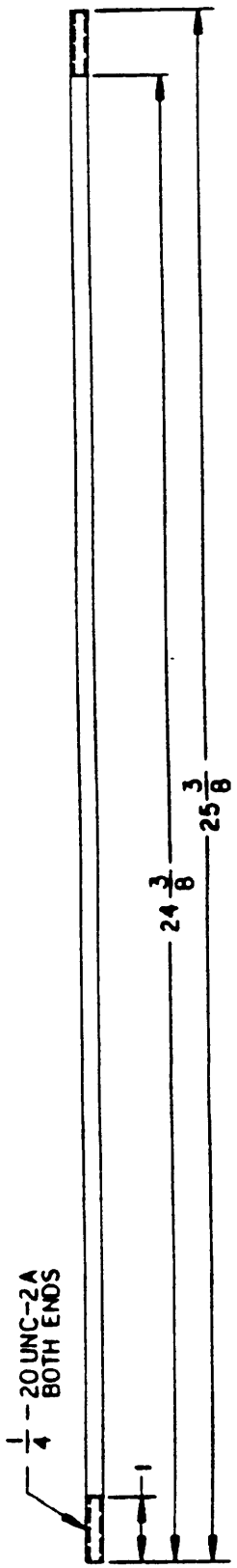
2 RIGHT-SIDE CHANNEL STEEL - HOT ROLLED 1 REQ'D.



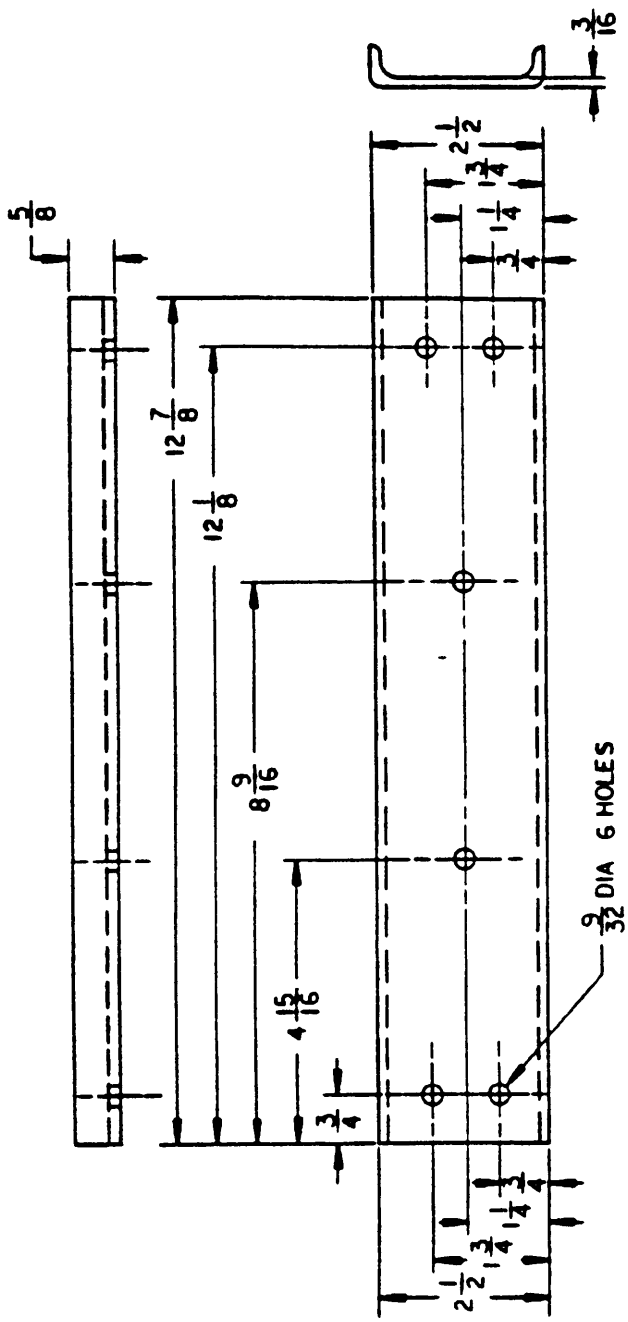
3 SCALE 1/32 THICK STAINLESS STEEL 1 REQ'D. (SEE NOTE 7

FIGURE 3. Main frame details (See Figure 8 for notes and bill of materials) - Continued.

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(4) GUIDE ROD
STEEL
2 REQ'D.



(5) TOP CHANNEL
STEEL - HOT ROLLED
1 REQ'D.

FIGURE 3. Main frame details (See figure 8 for notes and bill of materials) - Continued.

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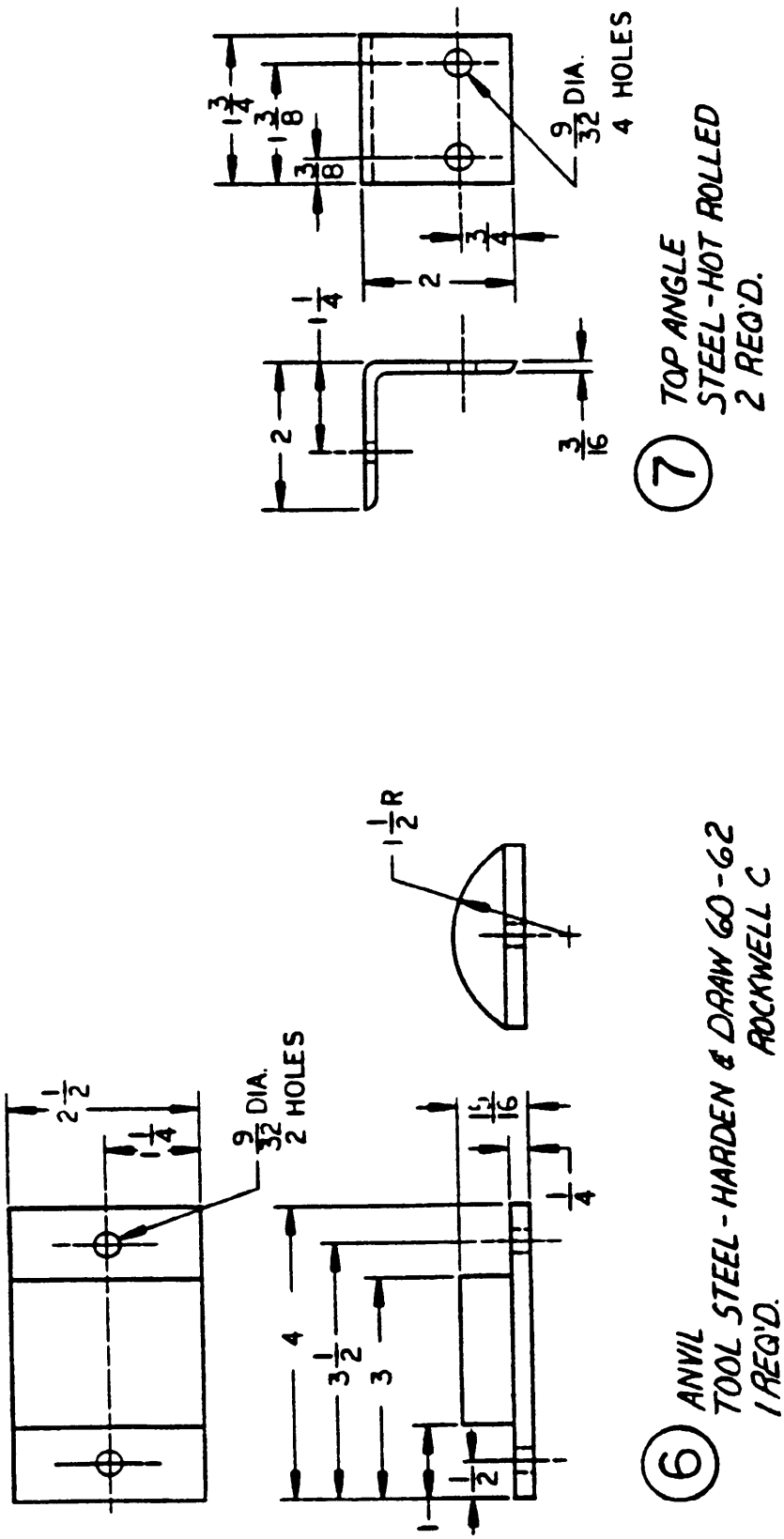


FIGURE 3. Main frame details (See figure 8 for notes and bill of materials) - Continued.

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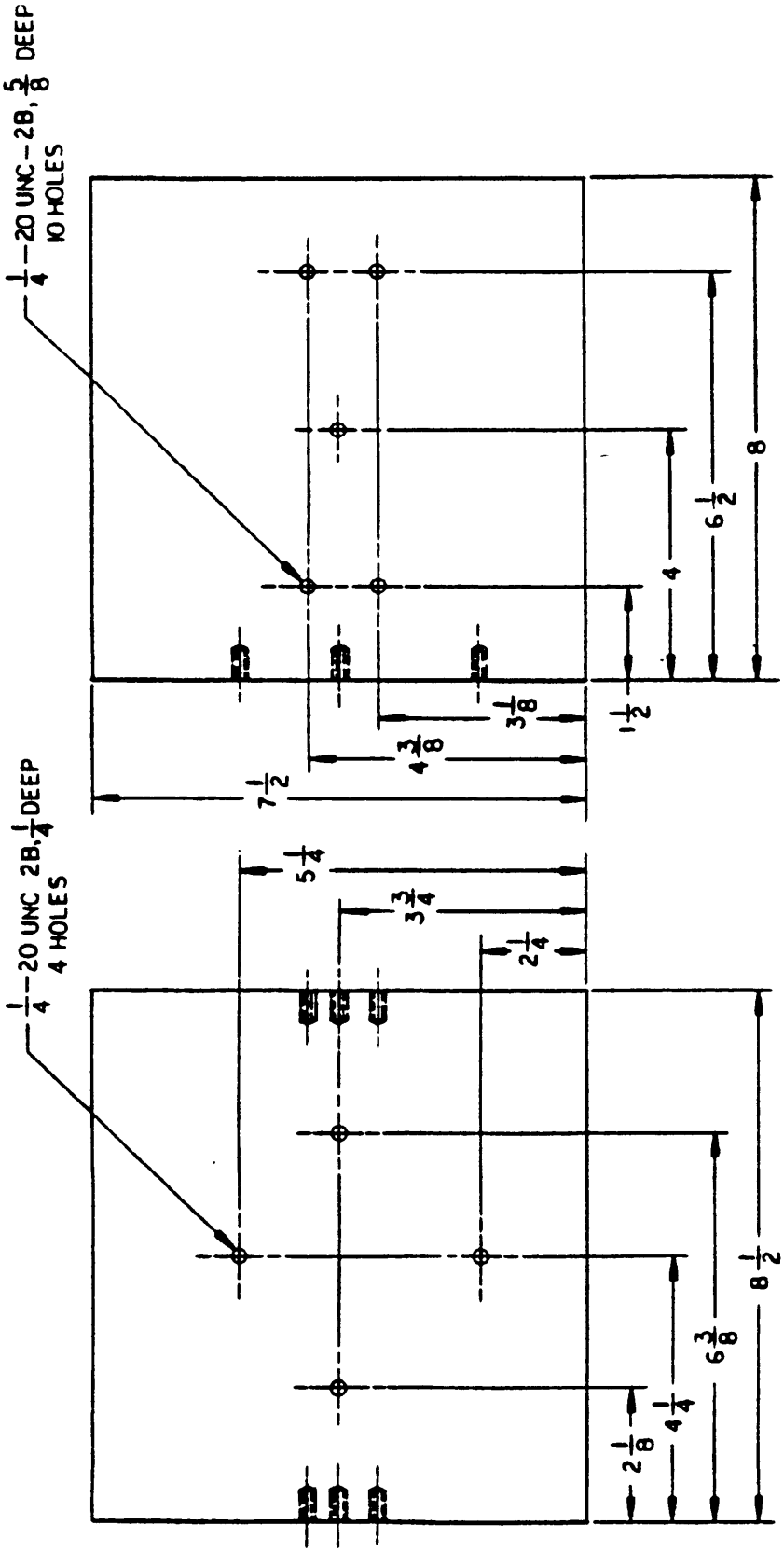


FIGURE 4. Carriage-assembly and base details (See figure 8 for notes and bill of materials).

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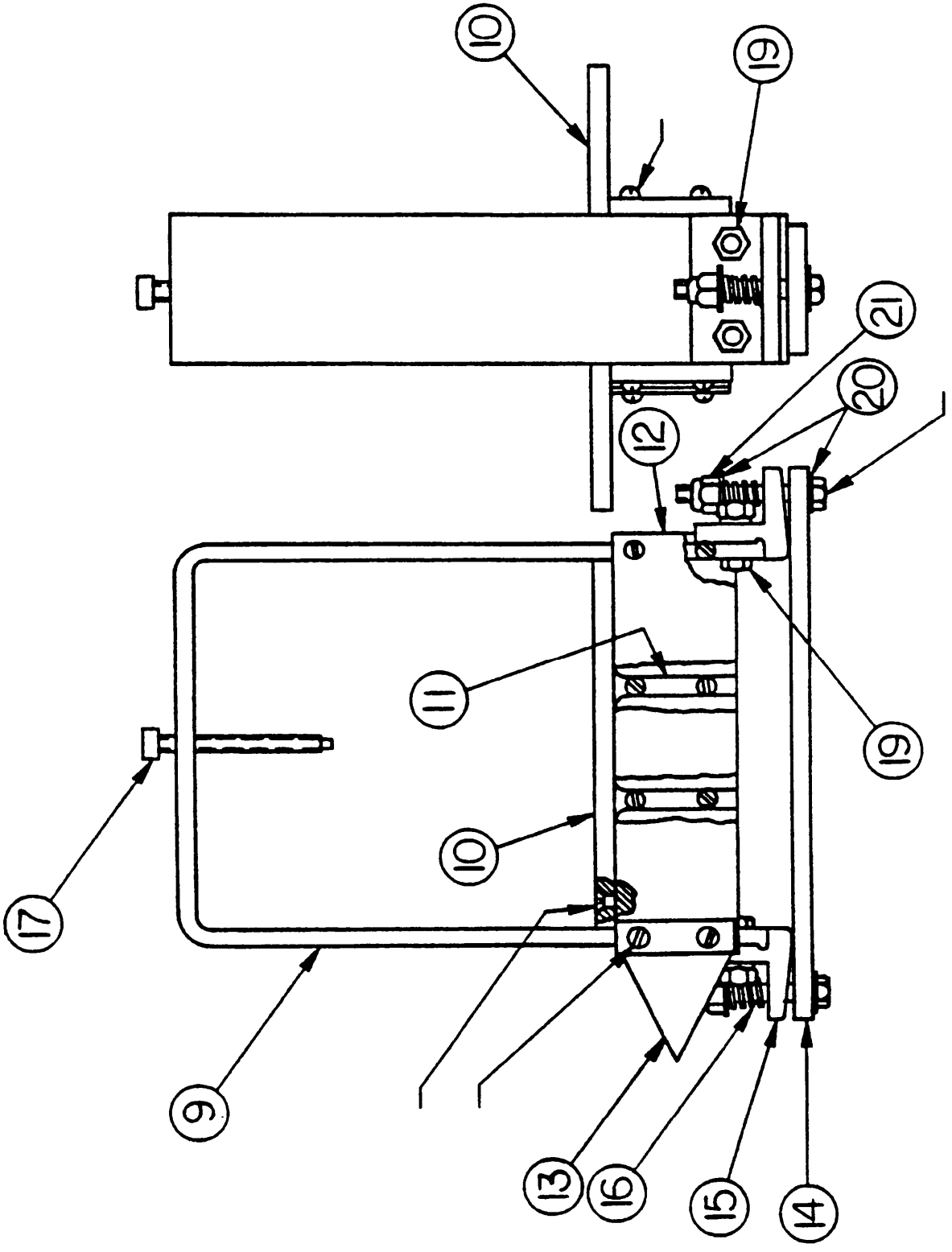
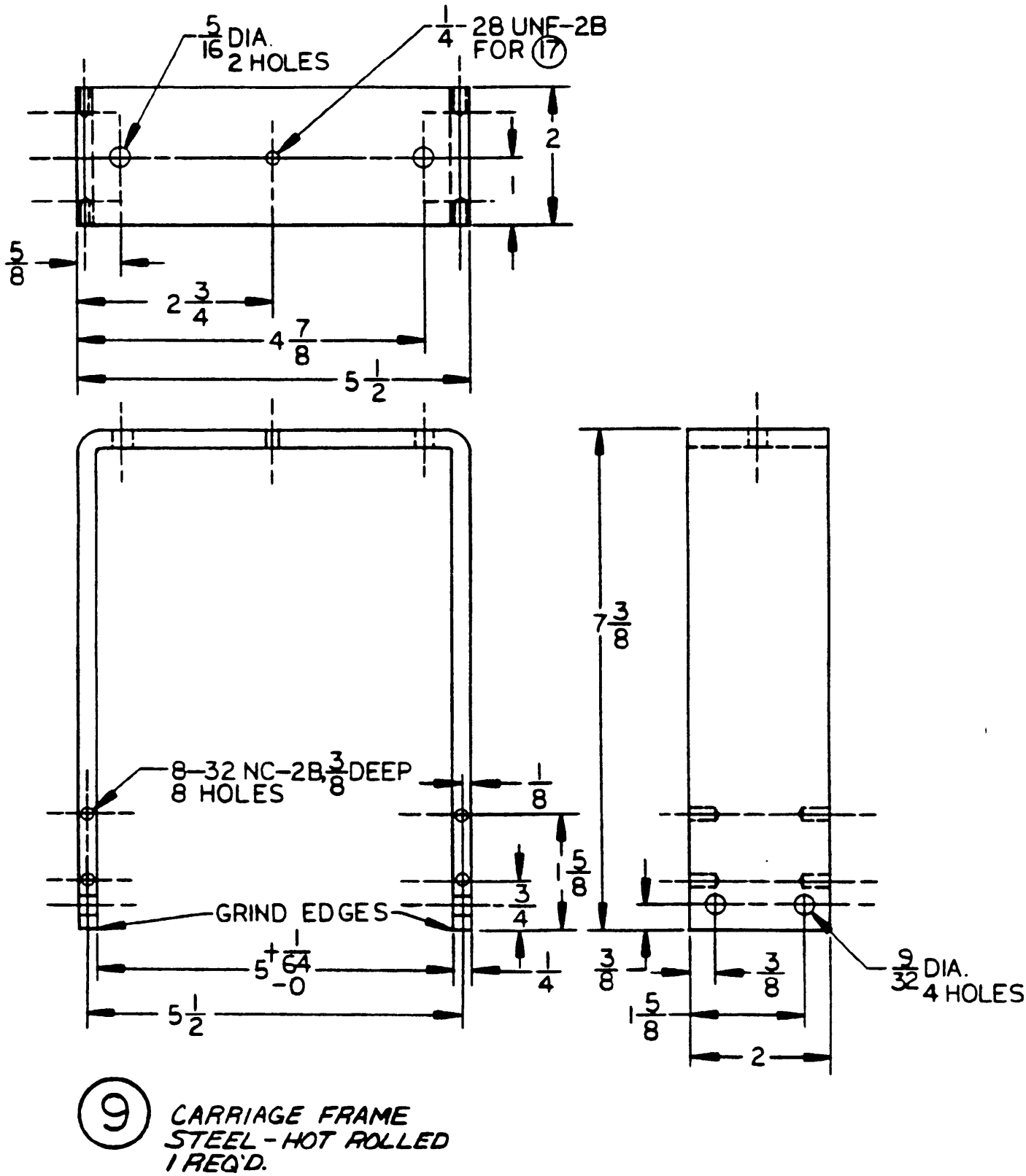


FIGURE 5. Carriage-assembly and base details (See figure 8 for notes and bill of materials).

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9 CARRIAGE FRAME
STEEL - HOT ROLLED
1 REQ'D.

FIGURE 6. Carriage details (See fig 8 for notes and bill of materials).

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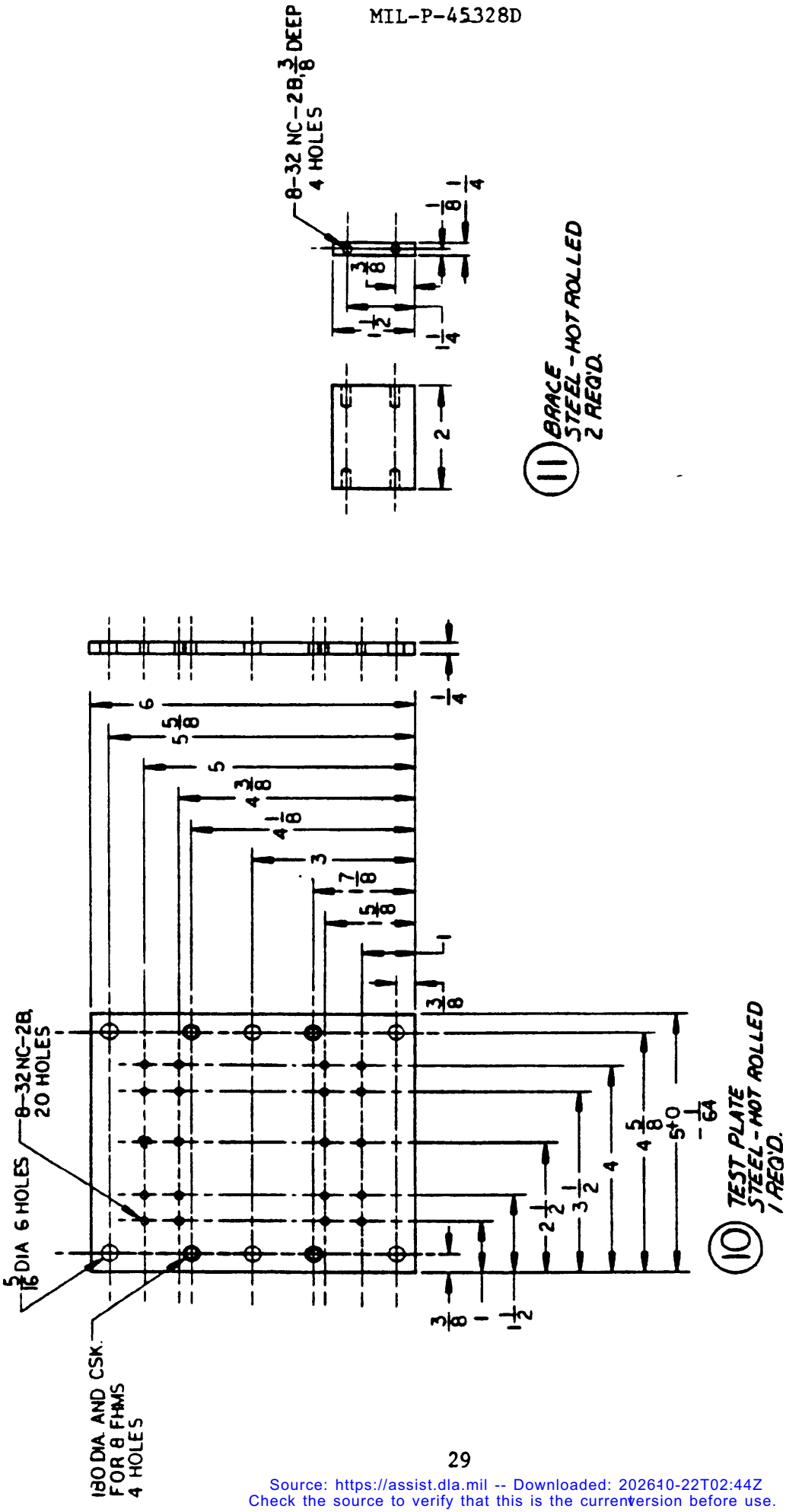
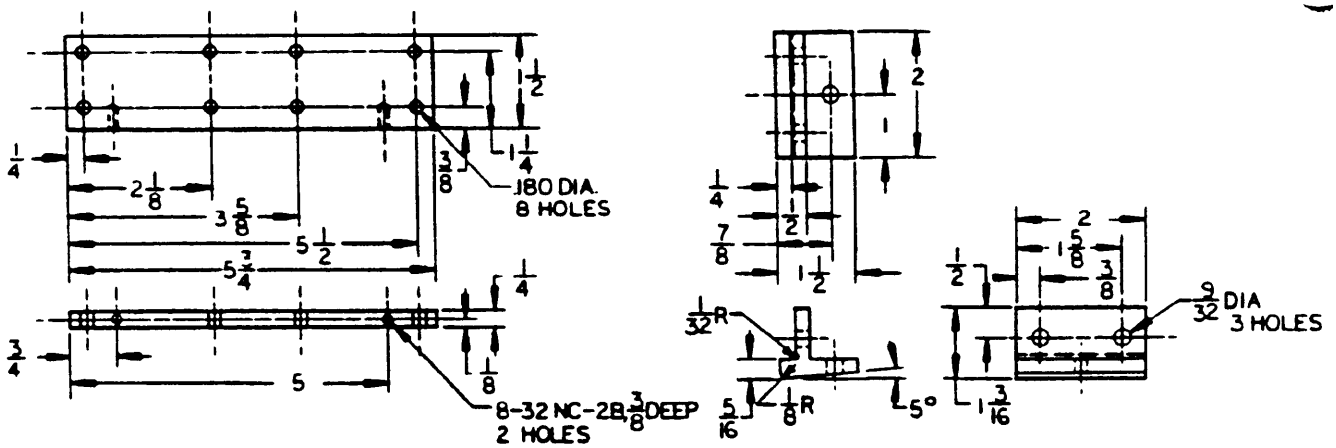


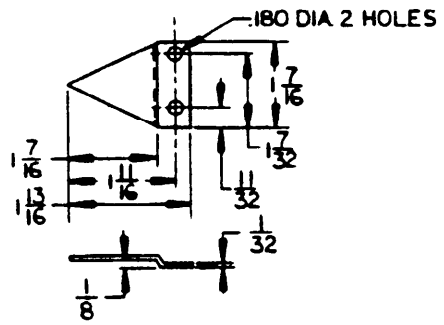
FIGURE 6. Carriage details (See figure 8 for notes and bill of materials)-Continued.

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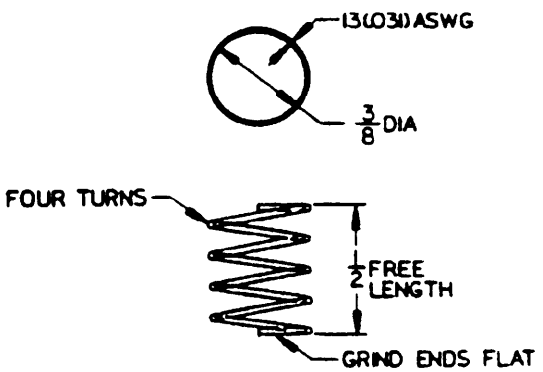


(12) SIDE PLATE
STEEL - HOT ROLLED
2 REQ'D.

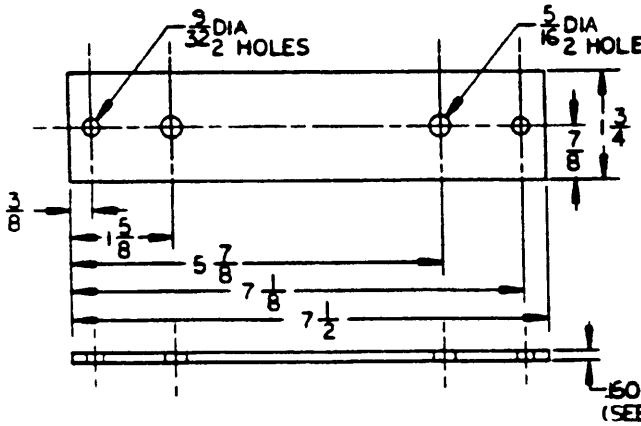
(15) PIVOT
TOOL STEEL - HARDEN & DRAW 60-62
ROCKWELL C
2 REQ'D.

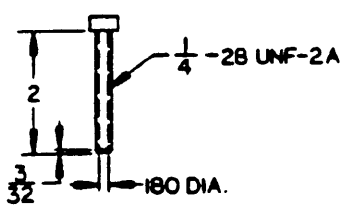


(13) POINTER
STEEL - STAINLESS
1 REQ'D.



(16) SPRING
MUSIC WIRE
2 REQ'D.





(17) HOLDING BOLT
STEEL
1 REQ'D.

(14) SPRING, CALIBRATED
TOOL STEEL (SEE NOTE 4)
1 REQ'D.

FIGURE 7. Carriage details (See figure 8 for notes and bill of material).

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Item No. or MS Part No.	Description	Quantity
1	Left-side channel	1
2	Right-side channel	1
3	Scale	1
4	Guide rod	2
5	Top channel	1
6	Anvil	1
7	Top angle	2
8	Base	1
9	Carriage frame	1
10	Test plate	1
11	Brace	2
12	Side plate	2
13	Pointer	1
14	Spring, calibrated	1
15	Pivot	2
16	Spring, compression	2
17	Squarehead bolt, No. 1/4-28 UNF-2A x 2 long, steel, modified	1
18	Acorn nut, No. 1/4-20 UNC-2B, steel	2
19	Hexagon nut, NO. 1/4-20 UNC-2B, steel	14
20	Flat washer, 1/4, steel	4
21	Stop nut, per note 5	2
MS35206-242	Screw	2
MS35206-244	Screw	16
MS35190-51	Screw	4
MS90725-6	Bolt	20
MS90725-7	Bolt	4
MS90725-12	Bolt	2
MS35338-4	Lockwasher	2
MS35338-6	Lockwasher	26

NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are 1/64 on fractions, .005 on decimals and l/2~ on angles.
3. Finish: The general assembly and the carriage shall be given a light spray coat of metal primer and then finished with a full spray coat of lusterless gray enamel. The mounting fixtures shall be cadmium or nickel plated. The spring, part 14, anvil, part 6, and pivots, part 15, shall be flash-dip nickeled.
4. Develop thickness experimentally so that spring constant will be 5,000 to 5,500 pounds-per-inch deflection when hardened and drawn.
5. Stop nut shall be catalog No. 29U040, as made by Elastic Stop Nut Corporation, Union, N.J., or equal.
6. The materials specified herein have been found satisfactory, but any suitable material may be used.
7. Scale shall be graduated in tenths of an inch. Fill block.

FIGURE 8. Notes and bill of materials.

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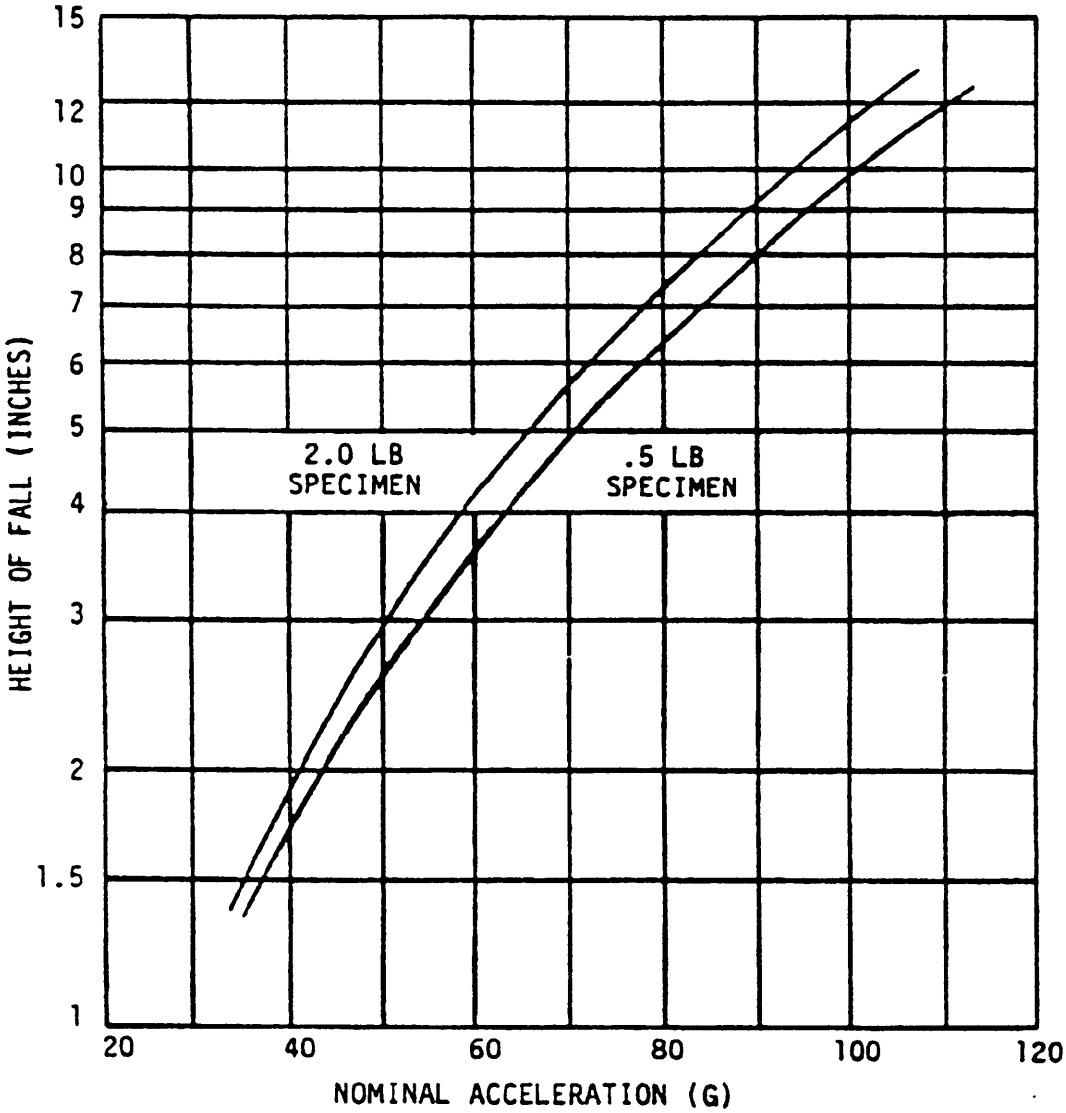


FIGURE 9. Typical calibration curves.

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Custodians:
Army-AT
Air Force - 99

Review activities:
DLA - CS
Army - ME
Air Force - 82

User activity:
Army-ME

Preparing activity:
Army - AT

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FSC 2910

DOCUMENT IDENTIFIER *(Number)* AND TITLE

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